GROUP PROJECT.

Title: Midland Energy Resources' Cost of Capital Projection for 2007

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Abstract

Our project estimates the cost of capital for Midland Energy Resources and its divisions, which are Exploration & Production, Refining & Marketing, and Petrochemicals. This aims to guide critical financial decisions in meeting the goals outlined by the management for the upcoming year. We choose to utilize the Weighted Average Cost of Capital (WACC) method for its adaptability in reflecting division-specific risks and financial structures. Key findings reveal Midland consolidated WACC of 8.13%, with division-specific estimates of 8.04% for E&P, 9.11% for R&M, and 5.92% for Petrochemicals. This highlights the varying capital costs driven by the operational and financial characteristics of each entity. The detailed calculations and analysis are documented to offer actionable insights for optimizing Midland's capital structure and investment decisions companywide.

1. Introduction

Midland Energy Resources is a capital intensive, global energy company with operations spanning oil and gas exploration and production, refining and marketing, and petrochemicals. The exploration and production division are the most profitable for the company, the refining and marketing division is the largest division of the company, while the petrochemicals division is the smallest.

In the near future, the company is actively expanding international presence through investing in foreign projects with local partners to access more resources and aim to reduce geopolitical risks. The company prioritizes valued projects across its divisions; Thus, Midland continuously uses advanced technologies to enhance productivity and efficiency. For financial strategy, Midland optimizes its capital structure to balance debt and equity to support shareholder value and stability. The management team also wants to perform stock repurchases to strengthen investors' confidence in the company's financial future.

Additionally, Midland will also maintain its position in the refining and petrochemical sectors by investing in facilities and expanding production capabilities. This commitment to innovation and market leadership will help sustain the company's competitive position in a dynamic global energy market.

The case study will determine the cost of capital for the parent company and its divisions in order to optimize its capital structure and make informed decisions. Various factors are to be considered, including market conditions, rates, future estimates, revenue, debt and equity values to find the appropriate cost of capital. To accurately calculate the Weighted Average Cost of Capital for Midland Energy Resources, we must also consider the company's investment and financial policies. risk factors and other estimates. This is because each division is unique and has its own debt ratio target.

2. Problem to Solve

In order to find the cost of capital, we would like to use the Weighted Average Cost of Capital (WACC) because it effectively captures the overall expense of financing the firm's operations through debt and equity. With the nature of WACC, it provides a comprehensive view on the average rate of return required by investors and debtholders. By weighing the costs of equity and debt proportionally to the firm value, WACC accurately reflects the total cost of financing Midland Energy Resources' operations and future projects.

Furthermore, Midland particularly has 3 distinct divisions: Exploration and Production, Refining and Marketing, and Petrochemicals. Each has a unique financial and operational characteristic. For instance, Exploration and Production is highly capital intensive, with $7.94 billion in capital expenditures in 2006 and substantial risk exposure due to geopolitical factors and the increasing reliance on deep-water drilling and natural gas. WACC accounts for these risks with the premium included in the cost of equity. Additionally, we would also estimate beta values based on comparable companies in the industry, ensuring they reflect the business risks of the sector. The cost of debt will also be adjusted for the market credit spreads required by creditors.

In contrast, the Refining and Marketing division operates in a commoditized market with small net margins, with $4 billion in after-tax earnings on $203 billion in revenue in 2006. Its relatively low debt ratio of 31% indicates a conservative borrowing approach, driven by the refining industry's competitive nature and low profit margins. This would leave little room for error in meeting debt obligations. Moreover, the high competition in this market makes it difficult to pass on cost to customers, resulting in unpredictable cash flows that would discourage the firm from financing business with a significant amount of debt. This approach minimizes solvency risk but limits the ability to amplify returns via the benefit of tax shield. Hence, WACC ensures that the distinct financial status of R&M division is appropriately reflected in capital allocation decisions.

The smallest division of the company is Petrochemicals, with $2.1 billion in after-tax earnings in 2006. This demonstrates a steady revenue stream and focuses on modernization and conducting joint ventures in growing international markets. The debt-to-equity ratio of 40 % and double A credit rating help this division to lower the cost of debt, which WACC could integrate to balance funding strategies against risk adjusted returns.

In general, the WACC model allows for adjustments at both the corporate and divisional levels. This enables precise decision making tailored to the unique risk return profiles of each division. For Exploration and Production industry, which has higher betas of 1.15 and faces greater commodity price volatility, using a divisional WACC ensures projects meet risk-adjusted rates. In contrast, for Refining and Marketing and Petrochemicals, which have lower risks but also lower margins, WACC helps avoid overinvesting in less profitable projects. WACC also

supports Midland's investment strategy by serving as a benchmark for evaluating profitable projects in the future with a realistic and sustainable cost of capital.

3. Problem Solving

a. Midland's Cost of Capital

As we have decided to use the weighted average cost of capital (WACC) basis for the company. We must find the cost of debt and cost of equity for Midland as required by WACC formula.

Based on Table 1, Midland (Consolidated) has a credit rating of A+, which corresponds to the spread of 1.62%. For this case, we decided to use the 10-years Treasury yield as the risk-free rate. We are convinced that the 10-year Treasury yield Table 2 of 4.66% is ideal as the risk-free rate for Midland's consolidated cost of capital because it balances long-term stability with relevance to corporate decision-making. The 1-year Treasury yield of 4.54 percent is too short-term and unsuitable for Midland's long-horizon investments, while the 30-year yield of 4.98% may overestimate risk due to its exposure to inflation and interest rate volatility over a long period. Hence, the cost of debt is as following:

Cost of Debt = Risk-free rate + Yield Spread
Cost of Debt = 4.66% +1.62%
Cost of Debt = **6.28%**

Before continuing the model, we have to find the corporate tax rate, which is not specified in the case description. From the income statement in exhibit 1, we would use the average tax rate of the previous 3 years as the predicted tax rate for 2007. The calculation is as following:

	2004	**2005**	**2006**
Income Before Taxes	17,910	32,723	30,447
Tax amount	7,414	12,830	11,747
Tax rate	41.4%	39.58%	38.58%
Average tax rate	39.73%		

In order to find the cost of equity, the first step that we need to take is unleveraging beta from comparable industry companies to remove the impact of their debt levels and convert it to illustrate solely Midland's business risk. This ensures that the comparison is fair, regardless of how much debt other companies use. Then, we releverage this beta using Midland's debt-to-equity ratio to reflect its actual risk based on the company specific capital structure. In this case, exhibit 5 indicates that industrial beta of Midland as 1.25, and industry D/E as 59.3%. Thus. the unlevered beta is as following:

Unlevered beta = Levered Beta / (1+(1-tax) * D/E)
Unlevered beta = 1.25 / (1+ (1- 0.3973) * 0.593)
Unlevered beta = **0.9209**

We would re-lever beta by using the Midland target D/V ratio as indicated on Table 1 of 42.2%. From this D/V ratio, we know that the E/V must be 57.8%. Thus, the target D/E ratio is 0.7301. Hence, the Re-levered beta is as following:

Re-levered beta = Unlevered beta * (1+ (1 – tax) * target D/E)
Re-levered beta = 0.9209 * (1+(1-0.3973) * 0.7301)
Re-levered beta = **1.3261**

The case specifies that Midland used the equity market risk premium of 5% in 2006. Although there have been several other figures that have been used in the past, we believed that 5% would be the appropriate figure for 2007. The reason is that we do not have enough information indicating possible changes in 2007, so reusing the previous figure would be reasonable in this case. With CAPM formula, the cost of equity would be as following:

Cost of Equity = Risk-free rate + beta * EMRP
Cost of Equity = 0.0466 + 1.3261 * 0.05
Cost of Equity = **11.29%**

Now, we can find the cost of capital for Midland using the WACC formula:

WACC – Midland = 6.28% * (1 – 39.73%) * 42.2% + 11.29% * 57.8%
WACC – Midland = 0.0628 * 0.6027 *0.422 + 0.0653
WACC – Midland = 0.016 + 0.0653 = **0.0813**

From the calculation above, we find that the Cost of Capital of Midland Energy Resources as of 2007 is 8.13%.

b. Exploration & Production division Cost of Capital

The E&P division could be the most expensive part of Midland Energy Resources to operate because it requires large investments and comes with higher risks. We calculated the division specific WACC to evaluate its unique cost of financing.

Cost of Debt

From *Table 1*, the E&P division has a credit rating of **A+**, which corresponds to a spread of **1.60%** over the risk-free rate. We use the 10-year Treasury yield of **4.66%** as the risk-free rate, as it reflects an appropriate balance between long-term stability and project horizons.

The choice of a 10-year yield aligns well with the extended timelines of Midland's exploration and production activities. The shorter 1-year yield of **4.54%** is too short-term, while the longer 30-year yield of **4.98%** would overstate risk due to interest rate volatility and inflation.

The cost of debt is calculated as follows:

$$\text{Cost of Debt} = \text{Risk-free rate} + \text{Yield Spread}$$

$$\text{Cost of Debt} = 4.66\% + 1.60\% = \mathbf{6.26\%}$$

Tax Rate

The corporate tax rate, based on *Exhibit 1*, is the average of the last three years:

$$\text{Average Tax Rate} = \mathbf{39.73\%}$$

Thus, the after-tax cost of debt is:

$$\text{After-Tax Cost of Debt} = \text{Cost of Debt} \times (1 - \text{Tax Rate})$$

$$\text{After-Tax Cost of Debt} = 6.26\% \times (1 - 0.3973) = \mathbf{3.77\%}$$

Cost of Equity

To estimate the cost of equity, we use the Capital Asset Pricing Model (CAPM):

$$\text{Cost of Equity} = \text{Risk-free rate} + \beta \times \text{EMRP}$$

The division's beta is adjusted to reflect its capital structure and business risk. From *Exhibit 5*, the industry average beta for comparable companies in the E&P sector is **1.15**. First, we unlever the beta to account for the impact of debt:

$$\text{Unlevered Beta} = \text{Levered Beta} / 1 + (1 - \text{Tax Rate}) \times (D/E)$$

$$\text{Unlevered Beta} = 1.15 / 1 + (1 - 0.3973) \times 0.398 = \mathbf{0.927}$$

Next, we re-lever the beta using the E&P division's target capital structure. From *Table 1*, the division's target Debt-to-Value (D/V) ratio is **46%**, and Equity-to-Value C(E/V) is **54%**, giving a Debt-to-Equity (D/E) ratio of 46/54 = **0.852** :

$$\text{Re-levered Beta} = \text{Unlevered Beta} \times (1 + (1 - \text{Tax Rate}) \times (D/E))$$

$$\text{Re-levered Beta} = 0.927 \times (1 + (1 - 0.3973) \times 0.852) = \mathbf{1.403}$$

Using an equity market risk premium (EMRP) of **5.0%**, as employed in the analysis of Midland's consolidated cost of capital in the previous section, the cost of equity for the Exploration & Production division is calculated as follows:

$$\text{Cost of Equity} = 4.66\% + 1.403 \times 5.0\% = \mathbf{11.67\%}$$

Weighted Average Cost of Capital (WACC)

Finally, we compute the WACC for the E&P division:

$$\text{WACC} = \text{After-Tax Cost of Debt} \times (D/V) + \text{Cost of Equity} \times (E/V)$$

$$\text{WACC} = 3.77\% \times 46\% + 11.67\% \times 54\%$$

$$\text{WACC} = 1.73\% + 6.3\% = \mathbf{8.04\%}$$

We thus determine that the Cost of Capital for the Exploration & Production Division of Midland Energy Resources is **8.04%** as of 2007. This reflects the division's higher business risk due to geopolitical exposure and reliance on deep-water drilling and liquefied natural gas investments.

c. Refining & Marketing division Cost of Capital

Cost of Debt:

From Table 1, the Refining & Marketing Division has a credit rating of BBB, which corresponds to a spread of **1.80%** over the risk-free rate. We use the 10-year Treasury yield of **4.66%** as the risk-free rate, as it reflects an appropriate balance between long-term stability and project horizons.

The choice of a 10-year yield aligns well with the extended timelines of Midland's Refining & Marketing Division's activities. The shorter 1-year yield of **4.54%** is too short-term, while the longer 30-year yield of **4.98%** would overstate risk due to interest rate volatility and inflation.

The cost of debt is calculated as follows:

$$\text{Cost of Debt} = \text{Risk-free rate} + \text{Yield Spread}$$

$$\text{Cost of Debt} = 4.66\% + 1.80\% = \mathbf{6.46\%}$$

Tax Rate:

The corporate tax rate, based on Exhibit 1, is the average of the last three years. As calculated previously:

$$\text{Average Tax Rate} = \mathbf{39.73\%}$$

Thus, the after-tax cost of debt is:

$$\text{After-Tax Cost of Debt} = \text{Cost of Debt} \times (1 - \text{Tax Rate})$$

$$\text{After-Tax Cost of Debt} = 6.46\% \times (1 - 0.3973) = \mathbf{3.89\%}$$

Cost of equity:

To estimate the cost of equity, we use the Capital Asset Pricing Model (CAPM):

$$\text{Cost of Equity} = \text{Risk-free rate} + \beta \times \text{EMRP}$$

The division's beta is adjusted to reflect its capital structure and business risk. From Exhibit 5, the industry average beta for comparable companies in the Refining & Marketing sector is 1.20. First, we unlever the beta to account for the impact of debt:

$$\text{Unlevered Beta} = \text{Levered Beta} / (1 + (1 - \text{Tax Rate}) \times (D/E))$$

$$\text{Unlevered Beta} = 1.20 / (1 + (1 - 0.3973) \times 0.203) = \mathbf{1.0692}$$

Next, we re-lever the beta using the Refining & Marketing division's target capital structure. From Table 1, the division's target Debt-to-Value (D/V) ratio is **31%**, and Equity-to-Value C(E/V) is **69%**, giving a Debt-to-Equity (D/E) ratio of 31/69 = **0.449**

$$\text{Re-levered Beta} = \text{Unlevered Beta} \times (1 + (1 - \text{Tax Rate}) \times (D/E))$$

$$\text{Re-levered Beta} = 1.0692 \times (1 + (1 - 0.3973) \times 0.449) = \mathbf{1.3585}$$

Using an equity market risk premium (EMRP) of 5.0%, as employed in the analysis of Midland's consolidated cost of capital in the previous section, the cost of equity for the Refining & Marketing division is calculated as follows:

$$\text{Cost of Equity} = \text{Risk-free rate} + (\beta \times \text{EMRP})$$

$$\text{Cost of Equity} = 4.66\% + (1.3585 \times 5.0\%) = \mathbf{11.45\%}$$

WACC

Finally, we compute the WACC for the Refining & Marketing division:

$$\text{WACC} = [\text{After-Tax Cost of Debt} \times (D/V)] + [\text{Cost of Equity} \times (E/V)]$$

$$\text{WACC} = (3.89\% \times 31\%) + (11.45\% \times 69\%)$$

$$WACC = 1.2059\% + 7.9\% = \mathbf{9.11\%}$$

We thus determine that the Cost of Capital for the Refining & Marketing division of Midland Energy Resources is 9.11% as of 2007.

d. Petrochemicals division Cost of Capital

The Petrochemicals division has a distinct financial profile with lower risk and steady revenue streams. To evaluate its cost of capital, we calculated its division specific WACC, reflecting its unique financial and operational characteristics.

Cost of debt

Based on Table 1, the Petrochemicals division has a credit rating of AA-, corresponding to a spread of 1.35% over the risk-free rate. We selected the 10-year Treasury yield of 4.66% as the risk-free rate, as it strikes an appropriate balance between stability and alignment with Midland's project horizons.

The cost of debt is calculated as follows:

$$\text{Cost of Debt} = \text{Risk-free rate} + \text{Yield Spread}$$

$$\text{Cost of Debt} = 4.66\% + 1.35\% = \mathbf{6.01\%}$$

Tax Rate

Next, we find the corporate tax rate, by calculating the average tax rate of the previous 3 years, assuming it is the tax rate for next year, i.e. 2007. The previous calculations suggest that the average tax rate is 39.73%.

We then calculate the After-tax cost of debt.

$$\text{After tax cost of debt} = \text{Cost of Debt} \times (1 - \text{Tax Rate})$$

$$\text{After-Tax Cost of Debt} = 6.01\% \times (1 - 0.3973) = \mathbf{3.62\%}$$

Cost of Equity

The next step in determining the cost of capital for the Petrochemicals division involves calculating the cost of equity using the CAPM. This model requires the risk-free rate, the equity market risk premium, and the division's beta, which measures its sensitivity to market risk.

To estimate the beta for the Petrochemicals division, we first calculate the weighted average of the unlevered betas of the Exploration & Production and Refining & Marketing divisions, along with the parent company's beta. Asset values from each division are used to determine their respective weights in the overall company:

- E&P Asset Value: $140,099.57

- R&M Asset Value: $93,828.81

- Petrochemicals Asset Value: $28,449.78

- Total Assets: $262,378.16

The weights for each division are calculated as follows:

$$\text{Weight (E\&P)} = 140{,}099.57 \div 262{,}378.16 = 0.53$$

$$\text{Weight (R\&M)} = 93{,}828.81 \div 262{,}378.16 = 0.36$$

$$\text{Weight (Petrochemicals)} = 28{,}449.78 \div 262{,}378.16 = 0.11$$

We then form the equation after we get all the required values

$$W(E\&P)*\beta + W(R\&M)*\beta + W(P)*\beta = \beta\ (\text{Midland})$$

$$(0.53)*(0.927) + (0.36)*(1.0692) + 0.11*\beta = 0.921$$

$$0.4913 + 0.3849 + 0.11*\beta = 0.921$$

$$\beta = (0.921 - 0.8762)/0.11 = \mathbf{0.4072}$$

Thus, the unleveraged beta of Petrochemical division is 0.4072.

Now, we relever this beta as following:

$$\text{Re-levered Beta} = \text{Unlevered Beta} \times (1 + (1 - \text{Tax Rate}) \times (D/E))$$

$$\text{Re-levered Beta} = 0.4072 \times (1 + (1 - 0.3973) \times 0.6667) = \mathbf{0.57}$$

Using a market risk premium of 5.0%, the cost of equity was calculated as:

$$\text{Cost of equity} = \text{Risk-free rate} + \beta*(\text{EMRP})$$

$$\text{Cost of equity} = 0.0466 + 0.57*(0.05) = 0.0745 = \mathbf{7.45\%}$$

Weighted Average Cost of Capital (WACC)

Finally, we computed the division's WACC using its target capital structure, with a debt-to-value ratio (D/V) of 40% and an equity-to-value ratio (E/V) of 60%:

$$\text{WACC} = \text{After-Tax Cost of Debt} \times (D/V) + \text{Cost of Equity} \times (E/V)$$

$$\text{WACC} = 3.62\% \times 40\% + 7.45\% \times 60\%$$

$$\text{WACC} = \mathbf{5.92\%}$$

We conclude that the Cost of Capital for the Petrochemicals Division of Midland Energy Resources is 5.92% as of 2007.

4. Conclusion

To sum up, the analysis highlights that the cost of capital for each division of Midland Energy Resources is closely linked to its unique operational and financial attributes. These results underscore the necessity of utilizing division specific WACC estimates to inform both investment strategies and financial decisions. The calculated WACC for each division, along with their respective credit ratings and target debt-to-value ratios, are detailed in the table below:

Division	**WACC**	**Credit Rating**	**Target Debt / Value**
Exploration and Production	8.04%	A+	46.00%
Refining and Marketing	9.11%	BBB	31.00%
Petrochemicals	5.92%	AA-	40.00%
Midland Energy (Consolidated)	8.13%	A+	42.20%

The Exploration and Production division, with a WACC of 8.04%, demonstrates the inherent high-risk and capital-intensive characteristics of its operations. This division's performance is heavily influenced by fluctuations in commodity prices, geopolitical uncertainties, and its dependency on deep-water drilling. The A+ credit rating and a target debt-to-value ratio of 46% signify the division's capability to absorb these risks while maintaining a stable financial foundation.

The Refining and Marketing division, with the highest WACC of 9.11%, reflects the challenges of operating in a highly competitive and commoditized market. Its lower credit rating of BBB and conservative target debt-to-value ratio of 31% underlines the division's constrained pricing power and susceptibility to market volatility. These conditions demand a cautious approach to leveraging debt to finance its operations.

On the other hand, the Petrochemicals division stands out with the lowest WACC of 5.92%. This is supported by consistent revenue generation, a strong AA- credit rating, and a moderate target debt-to-value ratio of 40%. These factors underscore the division's ability to mitigate risks and secure cost-effective financing, making it well-positioned for stable growth and strategic investments.

At the corporate level, Midland Energy Resources' consolidated WACC of 8.13% reflects a balanced financial structure and operational strategy across its divisions. This benchmark serves as a critical tool for assessing enterprise-level initiatives, establishing performance targets, and evaluating opportunities for mergers and acquisitions.

Recommendations:

To achieve its financial objectives, Midland Energy must deploy an investment strategy that aligns the cost of capital with the unique risk-return profiles of its divisions. It is decisive to ensure that projects in higher-risk divisions, such as E&P, are evaluated against their divisional WACC to prevent underestimating capital costs. Otherwise, there could be unprofitable investments in the long term. Conversely, lower-risk divisions, like Petrochemicals, should leverage their favorable cost of capital to support strategic modernization and expansion, enabling the division to achieve scalable returns with minimized financial risk.

We recommend Midland to revisit its cost of capital annually and in response to significant changes in business conditions or capital structure, such as large acquisitions, shifts in market conditions, or geopolitical issues. This regular assessment will enable the company to align its financial policies with evolving business needs, optimize its capital structure, and maintain a competitive edge in the energy sector.

Regular recalibration of these metrics will enable Midland to maintain optimal financial alignment. For example, any deviation from the E&P division's target debt-to-value ratio of 46% or the Petrochemicals division's 40% ratio may change our investment thresholds and funding strategies. By proactively aligning financial policies with updated cost-of-capital metrics, Midland Energy can strengthen its market position and secure long-term growth in a dynamic global energy landscape.

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